Ishaan Gupta

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Education

University of California San Diego **Sep 2023 – Mar 2028**
Ph.D. in Computer Science *La Jolla, CA*
- GPA 3.92 | Awarded **Charles Lee Powell Fellowship** for impactful research | M.S. awarded Jun 2025

University of California San Diego **Aug 2019 – Jun 2022**
B.S. in Computer Science, with Honors Distinction *La Jolla, CA*
- GPA 3.92 | Magna Cum Laude (Top 4%) | Vice President of Undergraduate Bioinformatics Club

Technical Skills

Software Dev	Java (Spring), Python (PyTorch, FastAPI), CUDA, C++, Microservices, SQL, AWS, MongoDB, Docker
Deep Learning	Transformers, CNN, LLMs, HuggingFace, Reinforcement Learning, PyTorch, scikit-learn, WandB
GenAI and NLP	LoRA, LangChain, SentenceTransformers, RAG, Agent Development Kit, MCP servers, Agent Tools

Research Experience

PhD Computer Science Researcher **Jan 2023 – Present**
UCSD Computer Science Department *San Diego, CA*
- Architected an LLM Finetuning pipeline using QLoRA, reasoning distillation, and reinforcement learning (GRPO) with multiple reward functions on *Qwen-8B*, beating *Opus 4.1* on genetic reasoning accuracy (61% vs 48%) at 50× lower latency.
- Developed DiploFormer, an end-to-end reproducible pipeline for adapting dilated CNN and transformer models to predict diseases from 1.5M DNA sequences, lifting correlations from 0.16 to 0.79 over the pretrained baseline. *ASHG* (2026).
- Invented state-of-the-art algorithm GenomeDecoder using approximate suffix matching and graph simplifications for high-throughput prediction of millions of bases long segmental duplications in DNA. *Bioinformatics* (2025).
- Optimized DNA sequence alignment algorithm (UniAligner) runtime by 10× using GPU acceleration and added decision trees to detect novel associations among human and primate immune genes. Published results in *Nature* (200+ citations).

Professional Experience

Intuitive Surgical, Inc. **Jun 2026 – Sep 2026**
ML Data Scientist Intern *Sunnyvale, CA*
- Built an AI pipeline for multimodal surgical data streaming from edge devices using 3D ConvNet to automate annotation.
- Engineered an agentic RAG interface for stakeholders, built on OCR + LLM ingestion of 1000+ documents into SQL and vector stores, routing queries to an intent-adaptive KPI (0.91 expert correlation) with forecasting and anomaly detection.
- Cut user-facing app load time by 3× through pre-aggregated schemas at multiple rollups, SQL indices, and multiprocessing.

Abterra Biosciences **Jul 2024 – Sep 2024**
Machine Learning Engineer Intern *San Diego, CA*
- Improved antibody seq precision 4× on millions of in-house samples by training Seq2Seq transformer PyTorch models.
- Trained LSTMs in TensorFlow for Entity Resolution achieving 95% accuracy (baseline 60%) using domain-specific features.
- Integrated DL models with FastAPI to serve production Java system, dropping the runtime 2 s to 0.1 s per sample.

Illumina, Inc. **Jun 2022 – Dec 2022**
Software Engineer I — Distributed Systems *San Diego, CA*
- Owned and extended 5 event-driven containerized Java Spring microservices for lab automation scanning 10,000+ samples/day, performing multi-step QC, and delivering real-time status updates using Amazon SQS to customers at <100 ms p95.
- Leveraged JDBC, SQL stored procedures, and RabbitMQ queues for routing protected data to 3 stateless ML services asynchronously and Spring Boot OAuth2 to secure client-facing REST APIs for real-time order history tracking.
- Eliminated a P0 Java concurrency race-condition bug impacting 1000+ customers, fixed and tested within 2 days.

Selected Projects

Graphazon — Graph-Augmented Semantic Product Recommendation [**ishaansd.github.io/Graphazon**]
Node2Vec · Knowledge Graphs · LambdaMART · NDCG
- Ranks most relevant products out of 300K+ using semantic similarity, LLM and KG embeddings, trained on Amazon ESCI.
- Shipped a live, rate-limited React UI over a containerized REST API on Google Cloud Run with cost-cap and LLM-call cache.

ChemBot — Agentic RAG Assistant for Trusted OTC Drug Information [**ishaansd.github.io/ChemBot**]
LangGraph · MCP · ChromaDB · Google Cloud Agent Platform
- Multi-agent retrieval of over-the-counter drug information (e.g., side effects and usage) grounded in trusted journals.

Publications & Conferences

- *DiploFormer: Deep Learning framework for capturing Non-Additive Regulatory Effects.* **ASHG**, 2026. [Github]
- *Complete sequencing of ape genomes.* **Nature**, 2025. [Paper] [Interactive plots]
- *GenomeDecoder: inferring segmental duplications in repetitive genomic regions.* **Bioinformatics**, 2025. [Paper]